Northeastern University

Northeastern University
Khoury College of Computer Sciences

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Computer Science Program

CS 5200 Database Management Systems

Midterm (Mock)

Friday, February 20th, 2026

Name ______________________________

ID ________________________

Duration: 90 minutes.

Question 1 10	
Question 2 10	
Question 3 10	
Question 4 10	
Question 5 10	
Question 6 10	
Question 7 20	
Question 8 20	
Total 100	

Instructions:

- ✓ *Check to ensure that you have a complete copy of the examination. There are **8** Questions, and a total of 17 (double-sided) pages. There should be no blank pages in your set.*
- ✓ *This examination is open book and open notes.*
- ✓ *Questions are <u>NOT</u> allowed during the exam.*
- ✓ *Make sure to solve all Questions so that you get partial credit even if you do not finish a question.*
- ✓ *Include all steps and calculations as appropriate for maximum credit. Be clear, brief, and specific in your answers.*
- ✓ *If you find a mistake correct it*
- ✓ *If anything is missing add it*

Good Luck!

Question #1

Draw the EER diagram to represent a small-private-airport database; the database is used to keep track of airplanes, their owners, airport employees, and pilots. The requirements for this database are as follows:

- Each AIRPLANE has a registration number [Reg#], is of a particular plane type [OF_TYPE], and is stored in a particular hangar [STORED_IN].
  - Each PLANE_TYPE has a model number [Model], a capacity [Capacity], and a weight [Weight].
  - Each HANGAR has a number [Number], a capacity [Capacity], and a location [Location].
- The database also tracks.
  - OWNERs of each plane [OWNS]
    - Each relationship instance in OWNS relates an AIRPLANE to an OWNER and includes the purchase date [Pdate].
  - the EMPLOYEEs who have maintained the plane [MAINTAIN].
    - Each relationship instance in MAINTAIN relates an EMPLOYEE to a service record [SERVICE].
- Each plane undergoes service many times. Each time the plane is serviced, a SERVICE record is added to the database and is linked to the plane.
  - A SERVICE record includes as attributes the date of maintenance [Date], the number of hours spent on the work [Hours], and the type of work done [Work_code].
  - The airplane registration number is used to identify a service record.
- An OWNER is either a person or a corporation.
- Both pilots [PILOT] and employees [EMPLOYEE] are PERSONs.
  - Each PILOT has attributes: license number [Lic_num] and restrictions [Restr]
  - Each EMPLOYEE has attributes salary [Salary] and shift worked [Shift].
  - All PERSONs in the database have data kept on their Social Security number [Ssn], name [Name], address [Address], and telephone number [Phone].
  - CORPORATIONs have attributes name [Name], address [Address], and telephone number [Phone].
- The database also keeps track of
  - the types of planes each pilot is authorized to fly.
  - the types of planes each employee can do maintenance work on

Answer to Question #1

Question #2

Consider the ER diagram for an airline reservations system in Figure 1. Extract from the ER diagram the requirements and constraints that produced this schema. Try to be as precise as possible in your requirements and constraints specification.

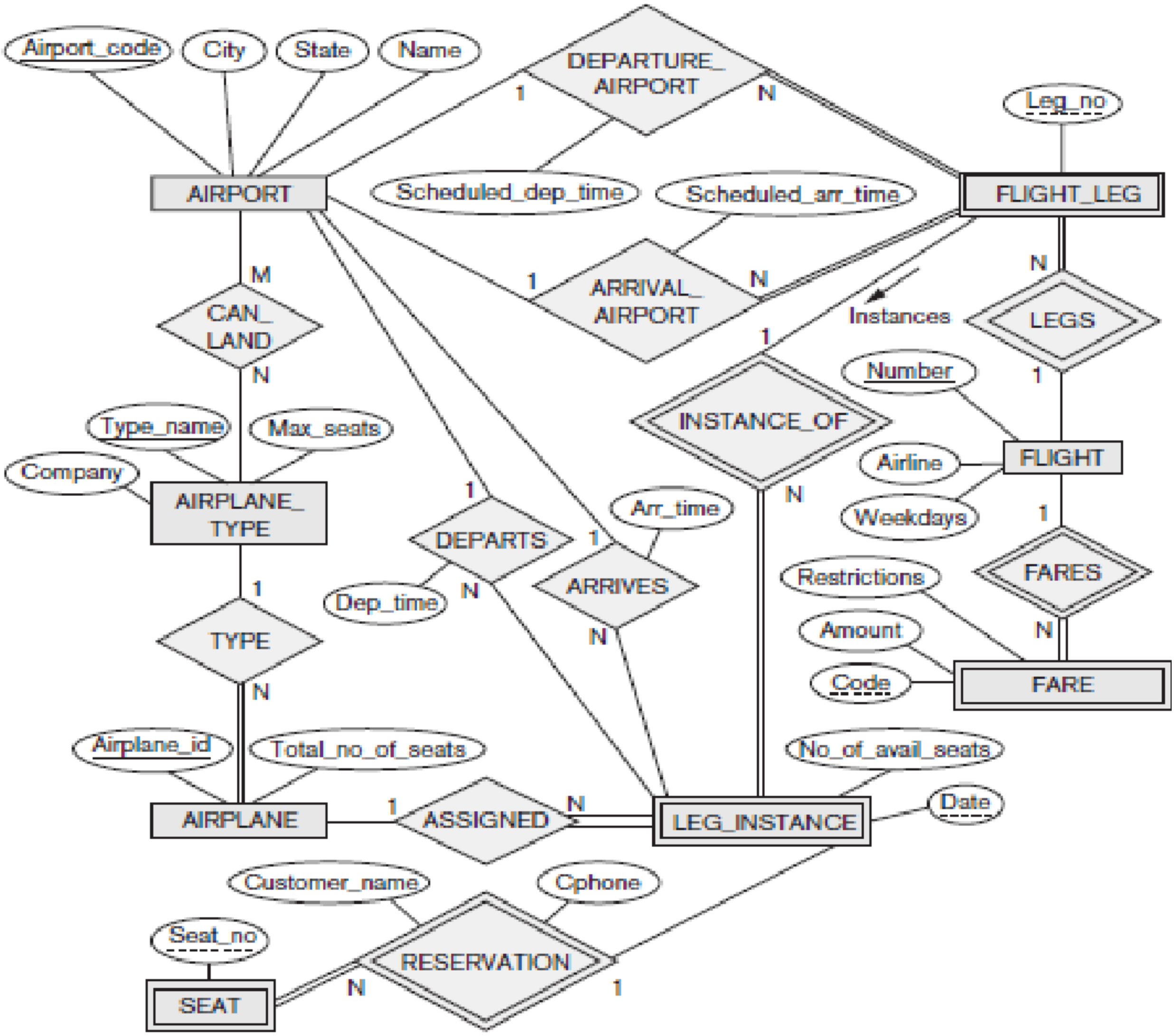

Figure 1: ER Diagram for Question 2

Answer to Question #2

Question #3

Consider the ER diagram in Figure 2. Assume that a course may or may not use a textbook, but that a text is a book that is used in some course. A course may not use more than five books. Instructors teach from two to four courses.

1. Supply (min, max) constraints on this diagram. *Clearly state any additional assumptions you make.*
2. If we add the relationship ADOPTS, to indicate the textbook(s) that an instructor uses for a course,
    a. should it be a binary relationship between INSTRUCTOR and TEXT, or a ternary relationship among all three entity types?
    b. What (min, max) constraints would you put on the relationship? Why?

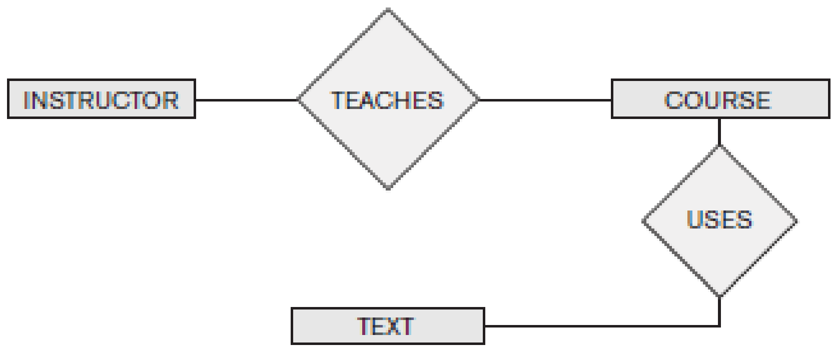

Figure 2: ER Diagram for Question 3

Answer to Question #3

Question #4

Consider the ER diagram in Figure 3. Use generalization to draw the corresponding EER Diagram

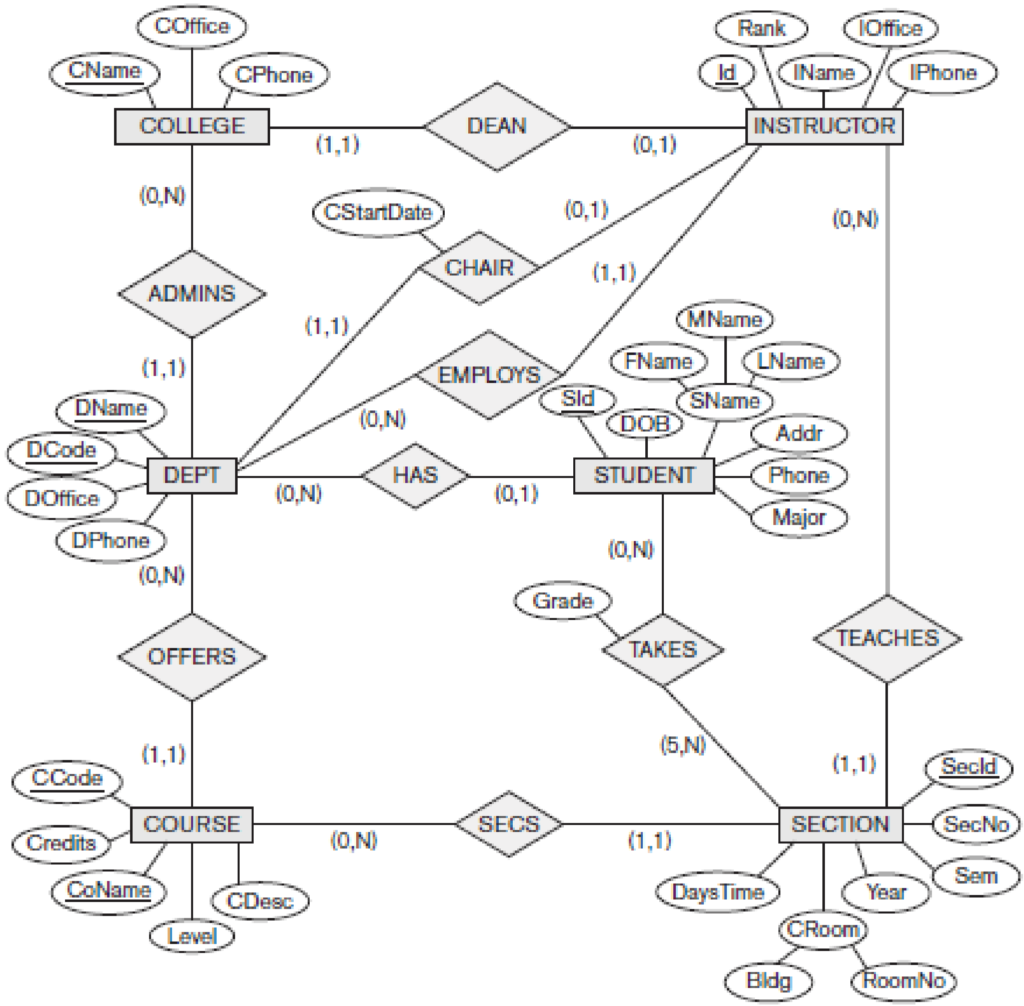

Figure 3: ER Diagram for Question 4

Answer to Question #4

Question #5

Write appropriate SQL DDL statements for declaring the LIBRARY relational database schema of Figure 4. Specify the keys and referential triggered actions.

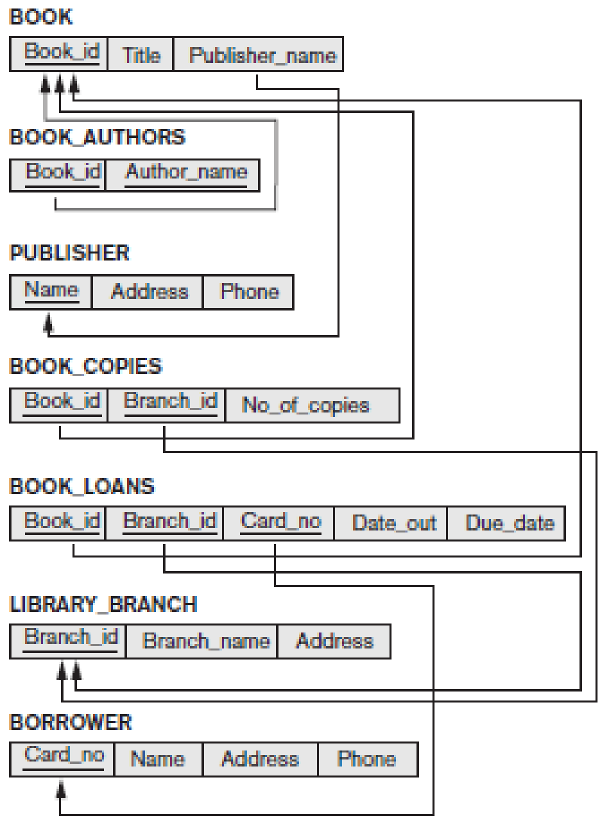

Figure 4 LIBRARY relational database schema for Question 5

Answer to Question #5

Question #6

Specify the following queries in SQL on the database schema of Figure 5.

1. Retrieve the names and major departments of all straight-A students (students who have a grade of A in all their courses).
2. Retrieve the names and major departments of all students who do not have a grade of A in any of their courses.

STUDENT

Name	Student_number	Class	Major
Smith	17	1	CS
Brown	8	2	CS

COURSE

Course_name	Course_number	Credit_hours	Department
Intro to Computer Science	CS1310	4	CS
Data Structures	CS3320	4	CS
Discrete Mathematics	MATH2410	3	MATH
Database	CS3380	3	CS

SECTION

Section_identifier	Course_number	Semester	Year	Instructor
85	MATH2410	Fall	07	King
92	CS1310	Fall	07	Anderson
102	CS3320	Spring	08	Knuth
112	MATH2410	Fall	08	Chang
119	CS1310	Fall	08	Anderson
135	CS3380	Fall	08	Stone

GRADE_REPORT

Student_number	Section_identifier	Grade
17	112	B
17	119	C
8	85	A
8	92	A
8	102	B
8	135	A

PREREQUISITE

Course_number	Prerequisite_number
CS3380	CS3320
CS3380	MATH2410
CS3320	CS1310

Figure 5 LIBRARY relational database schema for Question 5

Answer to Question #6

Question #7

Specify the following updates using the SQL update commands for the COMPANY database in Figure 6.

1. Insert <'Robert', 'F', 'Scott', '943775543', '1972-06-21', '2365 Newcastle Rd, Bellaire, TX', M, 58000, '888665555', 1> into EMPLOYEE.
2. Insert <'ProductA', 4, 'Bellaire', 2> into PROJECT.
3. Insert <'Production', 4, '943775543', '2007-10-01'> into DEPARTMENT.
4. Delete the WORKS_ON tuples with Essn = '333445555'.
5. Modify the Mgr_ssn and Mgr_start_date of the DEPARTMENT tuple with Dnumber = 5 to '123456789' and '2007-10-01', respectively.

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

DEPT_LOCATIONS

Dnumber	Dlocation
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

PROJECT

Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

Figure 6 Database of Question 7

Answer to Question #7

Question #8

Specify the following queries in SQL on the database in Figure 7.

1. Retrieve the names of all senior students majoring in 'CS' (computer science).
2. For each section taught by professor King, retrieve the course number, semester, year, and number of students who took the section.
3. Retrieve the names and major departments of all straight A students (students who have a grade of A in all their courses).
4. Insert a new student, <'Johnson', 25, 1, 'Math'>, in the database
5. Change the class of student 'Smith' to 2.

STUDENT

Name	Student_number	Class	Major
Smith	17	1	CS
Brown	8	2	CS

COURSE

Course_name	Course_number	Credit_hours	Department
Intro to Computer Science	CS1310	4	CS
Data Structures	CS3320	4	CS
Discrete Mathematics	MATH2410	3	MATH
Database	CS3380	3	CS

SECTION

Section_identifier	Course_number	Semester	Year	Instructor
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119	CS1310	Fall	08	Anderson
135	CS3380	Fall	08	Stone

GRADE_REPORT

Student_number	Section_identifier	Grade
17	112	B
17	119	C
8	85	A
8	92	A
8	102	B
8	135	A

PREREQUISITE

Course_number	Prerequisite_number
CS3380	CS3320
CS3380	MATH2410
CS3320	CS1310

Figure 7 Database of Question 8

Answer to Question #8